

Monitoring of Therapy

Therapeutic endpoints are evaluated by observation of clinical signs and symptoms, such as severity of pruritus and wheal number, size, and frequency. For drug toxicity, no specific monitoring is typically required beyond routine surveillance for adverse effects. However, certain individuals—such as those with impaired metabolism, co-morbid conditions, or those taking concomitant medications—may require closer monitoring and counseling regarding the use of H_1 antihistamines. Due to reports of hepatotoxicity, some sources recommend periodic liver transaminase evaluation when ciproheptadine is used.

Risks and Precautions

Adverse effects are common and vary by antihistamine class. Sedation is the most frequently reported side effect, particularly with first-generation H_1 antihistamines. The sedative effect is most pronounced with the ethanolamine and phenothiazine groups and less so with the alkylamine group. Sedation may diminish after several days of continuous use due to the development of tolerance. If sedation persists, switching to an agent from a different chemical group may be beneficial.

The use of H_1 antihistamines has been associated with an increased risk of occupational injuries and motor vehicle accidents. Other central nervous system (CNS) effects include dizziness, tinnitus, disturbed coordination, difficulty concentrating, blurred vision, and diplopia. Stimulatory CNS effects—such as nervousness, irritability, insomnia, and tremor—are more commonly seen with the alkylamine group.

Gastrointestinal side effects, including anorexia, nausea, vomiting, epigastric distress, diarrhea, and constipation, are frequently reported, especially with the ethylenediamine group. Taking the medication with food may help reduce these symptoms.

Anticholinergic effects include dry mucous membranes, urinary retention and hesitancy, postural hypotension, dizziness, erectile dysfunction, and constipation. These are most commonly associated with the ethanolamine, phenothiazine, and piperazine groups. Anticholinergic effects contraindicate the use of these agents in patients with narrow-angle glaucoma and necessitate careful monitoring in those with prostatic hypertrophy.

Cardiac toxicity is a serious concern, particularly with certain antihistamines. Arrhythmias, including QT interval prolongation and torsades de pointes, are dose-dependent and result from potassium channel blockade unrelated to H_1 receptor antagonism. Transient hypotension may occur following intravenous administration, especially if the drug is infused rapidly.

Cutaneous reactions following oral H_1 antihistamine use are rare but have been reported. These include eczematous dermatitis, allergic contact dermatitis, urticaria, petechiae, fixed drug eruptions, and photosensitivity.

Special Considerations

- The lowest effective dose should be used to minimize side effects.
- Dosage titration may be necessary after several days of therapy.
- Tolerance to sedation may allow for higher dosing in conditions like refractory chronic urticaria.
- Fexofenadine should not be taken with antacids due to reduced absorption.
- Patients with renal or hepatic impairment may require lower doses.
- Special populations (children, elderly, pregnant or breastfeeding women) may need individualized dosing.